An Internship Report

Of

Data Science Master Virtual Internship

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech (Computer Science and Engineering)

By

Roll Number of Student: 2410030850

Name of Student: Ankit Kumar

Batch: 2024-2028

1. CANDIDATE'S DECLARATION

I, **Ankit Kumar**, Roll No. **2410030850**, student of B.Tech in Computer Science and Engineering, Batch 2024–2028, IILM University, Greater Noida, hereby declare that this internship report titled **“Data Science Master Virtual Internship”** is prepared as a record of my learning and participation in the 8-week virtual internship conducted through the AICTE–EduSkills Virtual Internship Program and supported by Siemens during April–June 2026.

I have prepared this report to document the concepts, tools, methodologies and learning outcomes associated with the internship. The report is intended for academic submission and reflects the training and learning experience represented by my internship certificate.

I further declare that the information presented in this report has been compiled from the internship learning materials, my learning experience, the certificate issued to me, and publicly available program information used for understanding the internship structure.

Name	Ankit Kumar
Roll Number	2410030850
Programme	B.Tech – Computer Science and Engineering
Batch	2024–2028
Institute	IILM University, Greater Noida
Date	2026

2. ACKNOWLEDGEMENT

I would like to express my sincere gratitude to the AICTE–EduSkills Virtual Internship Program for providing me with the opportunity to participate in the Data Science Master Virtual Internship. The programme provided a structured environment for learning data science concepts and understanding how data-driven techniques can be applied to practical problems.

I am thankful to Siemens for supporting the internship and to EduSkills for facilitating the virtual learning experience. I also acknowledge the role of the AICTE and the National Internship Portal in creating opportunities for students to gain exposure to industry-oriented learning.

I would also like to thank IILM University, Greater Noida, and the faculty members of the Computer Science and Engineering department for their encouragement and academic support. Their guidance helped me remain consistent throughout the internship and connect the learning experience with my academic background.

Finally, I am grateful to everyone involved in providing the learning resources, assessments and technical content that helped me strengthen my understanding of Data Science, Data Engineering, Machine Learning and AI-related applications.

Regards,
Ankit Kumar
Roll No. 2410030850

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4. INTERNSHIP COMPLETION CERTIFICATE

The internship certificate supplied with this report confirms that **Ankit Kumar** of **IILM University, Greater Noida** successfully completed the **8-weeks Data Science Master Virtual Internship** during **April–June 2026**. The certificate identifies the programme as an AICTE–EduSkills Virtual Internship and shows Siemens as the supporting organization.

The original certificate is reproduced as the final appendix page of this report. The certificate page in the uploaded file is a one-page official credential containing the student's name, institute, internship title, duration and certificate identifiers.

5. PROJECT DESCRIPTION

5.1 Introduction

Data Science combines data preparation, statistical reasoning, machine learning, visualization and domain knowledge to extract useful information from data. The Data Science Master Virtual Internship provided a structured learning pathway around these areas, with emphasis on practical data workflows and industry-oriented problem solving.

The internship was completed virtually over eight weeks. The learning experience covered the data science lifecycle from identifying a problem and preparing data to building analytical or machine learning solutions, evaluating results and understanding deployment and platform-level considerations.

5.2 Organization / Programme Profile

EduSkills conducts AICTE-supported virtual internship programmes intended to provide students with structured industry-oriented learning. The AICTE National Internship Portal lists the Data Science Master Virtual Internship as a virtual programme with a duration of two months. The internship certificate provided by the student records an eight-week Data Science Master Virtual Internship during April–June 2026, supported by Siemens.

The programme is associated with the Altair/RapidMiner data science ecosystem. Learning areas commonly associated with the track include Data Engineering, Machine Learning, Applications & Use Cases and Platform Administration.

5.3 Problem Statement

Modern organizations generate large amounts of structured and unstructured data. A central challenge is to convert this raw information into reliable, useful and actionable insights. This requires a sequence of tasks including data access, cleaning, transformation, feature preparation, model building, evaluation and, where appropriate, deployment.

The internship therefore addressed the broader problem of developing an end-to-end understanding of data-driven problem solving. The learning focus was not limited to a single dataset; instead, it covered reusable workflows that can be applied to different data science and machine learning use cases.

5.4 Project / Internship Objectives

• Understand the fundamentals and workflow of Data Science and Machine Learning.

• Learn how data can be accessed, cleaned, transformed and prepared for analysis.

• Understand data engineering concepts and reusable data-processing workflows.

• Study classification, regression, model evaluation and related machine learning techniques.

• Explore feature engineering, model optimization and advanced analytical workflows.

• Understand business-oriented AI applications and the relationship between business problems and ML solutions.

• Gain awareness of model deployment, monitoring, scoring and operationalization.

• Understand the role of enterprise platforms such as RapidMiner AI Hub in managing data science processes.

• Develop analytical thinking, problem-solving ability and confidence in applying data-driven methods.

5.5 Scope of the Internship

The scope of the internship extended across the major stages of a data science lifecycle. At the data layer, the learning involved accessing, loading, cleaning, transforming and combining data. At the modeling layer, the focus included machine learning concepts, model selection, validation and optimization. At the application layer, the programme introduced deployment, monitoring, reporting and real-world use cases. Platform-oriented learning provided an overview of how enterprise data science environments can be administered and operated.

5.6 Technologies and Tools Used

Area	Technologies / Concepts
Data Science	Data analysis, data preparation, visualization, predictive analytics
Data Engineering	Data access, transformation, cleansing, joins, automation, text processing
Machine Learning	Classification, regression, clustering, feature engineering, model evaluation
Advanced ML	Ensemble methods, parameter optimization, time-series concepts, model selection
Applications	CRISP-DM, deployment, monitoring, scoring, dashboards and web-oriented use cases
Platform	RapidMiner / Altair ecosystem, RapidMiner Studio, AI Hub concepts
Programming / Integration	Python / R concepts, scripting and API-oriented workflows

These technologies and concepts represent the programme's data science workflow rather than a claim that every tool was used to build a single standalone software product. The internship was primarily a structured learning and assessment programme.

5.7 System Architecture / Data Science Workflow

A general workflow representing the internship's end-to-end data science perspective is shown below. It connects business understanding and data access with preparation, modeling, evaluation and deployment.

Problem / Business Need
↓
Data Access & Collection
↓
Data Cleaning & Preparation
↓
Feature Engineering / Transformation
↓
Machine Learning / Analytics
↓
Validation & Model Selection
↓
Deployment / Scoring / Monitoring
↓
Insights, Reporting & Decision Support

The workflow illustrates how individual learning modules fit together. Data engineering supports reliable inputs; machine learning converts prepared data into predictive or analytical models; applications and platform concepts help move those models toward operational use.

5.8 Methodology

Step 1 – Problem Understanding: Identify the objective and determine what type of data-driven solution is appropriate.

Step 2 – Data Access: Obtain relevant datasets and understand their structure, data types and sources.

Step 3 – Data Preparation: Clean inconsistent records, transform fields, handle missing or unsuitable values and prepare data for analysis.

Step 4 – Feature Engineering: Create or transform useful variables so that machine learning models can capture meaningful patterns.

Step 5 – Model Development: Apply appropriate supervised or unsupervised learning techniques according to the problem.

Step 6 – Evaluation: Validate model performance using suitable evaluation measures and compare candidate approaches.

Step 7 – Optimization: Improve the workflow through parameter tuning, model selection and appropriate feature choices.

Step 8 – Operationalization: Understand deployment, scoring, monitoring, dashboards and platform administration concepts.

5.9 Expected Learning Outcomes

• A stronger conceptual understanding of the complete data science lifecycle.

• Improved ability to reason about data quality, preparation and transformation.

• Familiarity with machine learning workflows and model evaluation.

• Awareness of advanced modeling, optimization and feature engineering techniques.

• Understanding of business-oriented AI use cases and CRISP-DM-style thinking.

• Awareness of deployment, monitoring, real-time scoring and enterprise platform operations.

• Improved problem-solving, analytical thinking and technical communication skills.

5.10 Certificates of Completion and Learning Evidence

The uploaded certificate provides direct evidence of completion. It states that Ankit Kumar, IILM University, Greater Noida, successfully completed the 8-weeks Data Science Master Virtual Internship during April–June 2026. The certificate also identifies the programme as an AICTE–EduSkills Virtual Internship and displays Siemens as the supporting organization.

The original certificate is included at the end of this report without alteration so that the academic submission contains the credential supplied by the student.

6. CONCLUSION

The Data Science Master Virtual Internship provided a structured introduction to the end-to-end data science lifecycle. The internship strengthened my understanding of data preparation, data engineering, machine learning, model evaluation, business-oriented AI applications and the operational side of data science platforms.

The experience also helped connect theoretical computer science knowledge with practical data-driven workflows. The exposure to tools and concepts such as RapidMiner, data preparation, machine learning, feature engineering, model optimization, deployment and platform administration provides a foundation for further learning and for future academic and industry projects.

Overall, the internship was a valuable learning experience that improved my technical awareness, analytical thinking and understanding of how data can be transformed into useful insights and deployable solutions.

7. BIBLIOGRAPHY / REFERENCES

1. AICTE National Internship Portal – Data Science Master Virtual Internship listing.

2. EduSkills Foundation – AICTE–EduSkills Virtual Internship Programme information.

3. Altair / RapidMiner learning resources and platform documentation.

4. Internship certificate supplied by Ankit Kumar, covering the April–June 2026 internship period.

5. Public programme and learning descriptions used to understand the Data Science Master curriculum.

Note: This report is prepared as an academic internship report based on the supplied certificate and published programme information. It does not claim completion of a separate commercial project unless such work is explicitly documented by the student.

8. INTERNSHIP CERTIFICATE – APPENDIX

शिक्षा मंत्रालय
MINISTRY OF
EDUCATION

सत्यमेव जयते

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All India Council for Technical Education

NATIONAL INTERNSHIP PORTAL

EduSkills
Nation Building Through Skills

AICTE - EduSkills
2026
VIRTUAL INTERNSHIP

Certificate of Virtual Internship

This is to certify that

Ankit kumar

IILM University, Greater Noida

has successfully completed the 8-weeks

Data Science Master Virtual Internship

During April - June 2026

May this Internship learning propel you toward a bright and successful career.

Supported By

SIEMENS

Gaurav Pai Head of Operations & Strategy SIEMENS	Dr. Buddha Chandrasekhar Chief Coordinating Officer (CCO) AICTE, Ministry of Education	Shubhajit Jagadev Chief Executive Officer (CEO) EduSkills

Certificate ID : 4ee8ae25393d7bedd6c4

Student ID: STU69e08784f0dc21776322436

INTERNSHIP GRADE
O

GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30